

NUMERICAL METHODS

Unit-3: Interpolation and Finite Differences

Comprehensive Study Notes



3.1 Introduction to Interpolation

Interpolation is a method of constructing new data points within the range of a discrete set of known data points. It is used to estimate values between known values.

Interpolation vs Extrapolation

Feature	Interpolation	Extrapolation
Definition	Estimating within known range	Estimating outside known range
Reliability	More reliable	Less reliable
Use	Filling gaps	Predicting future trends

3.2 Finite Differences and Operators

Finite differences are used to approximate derivatives and in interpolation formulas.

Forward Difference Operator (Δ)

$$\Delta f(x) = f(x + h) - f(x)$$

where h is the step size.

Higher Order Differences

$$\Delta^2 f(x) = \Delta f(x + h) - \Delta f(x) = f(x + 2h) - 2f(x + h) + f(x)$$

$$\Delta^3 f(x) = \Delta^2 f(x + h) - \Delta^2 f(x)$$

Backward Difference Operator (∇)

$$\nabla f(x) = f(x) - f(x - h)$$

Higher Order Differences

$$\nabla^2 f(x) = \nabla f(x) - \nabla f(x - h) = f(x) - 2f(x - h) + f(x - 2h)$$

Shift Operator (E)

$$E f(x) = f(x + h)$$

Properties:

- $E^2 f(x) = f(x + 2h)$
- $E^{-1} f(x) = f(x - h)$
- $E^n f(x) = f(x + nh)$

Relationship Between Operators

Operator	Relationship
E	$E = 1 + \Delta$
Δ	$\Delta = E - 1$
∇	$\nabla = 1 - E^{-1}$
Δ^2	$\Delta^2 = E^2 - 2E + 1$

Difference Table

Forward Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0				
		Δy_0			
x_1	y_1		$\Delta^2 y_0$		
		Δy_1		$\Delta^3 y_0$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$
		Δy_2		$\Delta^3 y_1$	
x_3	y_3		$\Delta^2 y_2$		
		Δy_3			
x_4	y_4				

Example: Difference Table for $y = x^2$

Given $y = x^2$ for $x = 0, 1, 2, 3, 4$:

x	$y = x^2$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	0				
		1			
1	1		2		
		3		0	

2	4		2		0
		5		0	
3	9		2		
		7			
4	16				

Observations:

- First differences: 1, 3, 5, 7 (arithmetic progression)
- Second differences: 2, 2, 2 (constant)
- Higher differences: 0

3.3 Newton Forward Interpolation Formula

Used when the point to be interpolated is near the **beginning** of the data set.

Formula

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

where:

$$p = (x - x_0) / h$$

Example: Find y at x = 1.2

x	0	1	2	3
y	1	2	5	10

Step 1: Construct difference table

x | y | Δy | $\Delta^2 y$ | $\Delta^3 y$

-----+-----				
0	1			
		1		
1	2		2	
		3		0
2	5		2	
		5		
3	10			

Step 2: Calculate p

$$p = (1.2 - 0) / 1 = 1.2$$

Step 3: Apply formula

$$\begin{aligned}y &= y_0 + p\Delta y_0 + [p(p-1)/2]\Delta^2 y_0 + [p(p-1)(p-2)/6]\Delta^3 y_0 \\y &= 1 + 1.2(1) + [1.2(0.2)/2]x^2 + [1.2(0.2)(-0.8)/6]x^0 \\y &= 1 + 1.2 + 0.24 + 0 \\y &= 2.44\end{aligned}$$

3.4 Newton Backward Interpolation Formula

Used when the point to be interpolated is near the **end** of the data set.

Formula

$$y = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

where:

$$p = (x - x_n) / h$$

Example: Find y at x = 1.8

x	0	1	2	3
y	1	2	5	10

Step 1: Backward difference table

x | y | ∇y | $\nabla^2 y$ | $\nabla^3 y$

0	1			
		1		
1	2		2	
		3		0
2	5		2	
		5		
3	10			

Step 2: Calculate p

$$p = (1.8 - 3) / 1 = -1.2$$

Step 3: Apply formula

$$y = y_n + p\nabla y_n + [p(p+1)/2]\nabla^2 y_n + [p(p+1)(p+2)/6]\nabla^3 y_n$$

$$y = 10 + (-1.2)(5) + [(-1.2)(-0.2)/2] \times 2 + [(-1.2)(-0.2)(0.8)/6] \times 0$$

$$y = 10 - 6 + 0.24 + 0$$

$$y = 4.24$$

Newton Forward vs Newton Backward

Feature	Newton Forward	Newton Backward
Best for	Beginning of data	End of data
p value	$p = (x - x_0)/h$	$p = (x - x_n)/h$
Term pattern	$p(p-1)(p-2)\dots$	$p(p+1)(p+2)\dots$

3.5 Gauss Forward Interpolation Formula

Used when the point lies **near the middle** (after center) of the data set.

Formula

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-1} + \dots$$

$$+ \frac{(p+1)p(p-1)(p-2)}{4!} \Delta^4 y_{-2} + \dots$$

Example: Find y at x = 2.5

x	0	1	2	3	4
y	1	3	7	13	21

Step 1: Difference table (center at x = 2)

x | y | Δy | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$

0	1				
		2			
1	3		2		
		4		0	
2	7		2		0
		6		0	
3	13		2		
		8			
4	21				

Center at $x_0 = 2$ ($y_0 = 7$)

$\Delta y_0 = 6$, $\Delta^2 y_{-1} = 2$, $\Delta^3 y_{-1} = 0$

Step 2: Calculate p

$p = (2.5 - 2) / 1 = 0.5$

Step 3: Apply formula

$$y = y_0 + p\Delta y_0 + [p(p-1)/2]\Delta^2 y_{-1} + \dots$$

$$y = 7 + 0.5(6) + [0.5(-0.5)/2] \times 2 + 0$$

$$y = 7 + 3 + (-0.25)$$

$$y = 9.75$$

3.6 Gauss Backward Interpolation Formula

Used when the point lies **near the middle** (before center) of the data set.

Formula

$$y = y_0 + p\Delta y_{-1} + (p+1)p/2! \Delta^2 y_{-1} + (p+1)p(p-1)/3! \Delta^3 y_{-2} + \dots$$

Example: Find y at x = 1.5

Same data as above:

Center at $x_0 = 2$ ($y_0 = 7$)

$$p = (1.5 - 2) / 1 = -0.5$$

Difference values:

$$y_0 = 7, \Delta y_{-1} = 4, \Delta^2 y_{-1} = 2, \Delta^3 y_{-2} = 0$$

Apply formula:

$$y = y_0 + p\Delta y_{-1} + [(p+1)p/2]\Delta^2 y_{-1} + \dots$$

$$y = 7 + (-0.5)(4) + [(0.5)(-0.5)/2] \times 2 + 0$$

$$y = 7 - 2 - 0.25$$

$$y = 4.75$$

Gauss Forward vs Gauss Backward

Feature	Gauss Forward	Gauss Backward
Position	After center	Before center

Formula terms	Uses $p, p-1, p-2...$	Uses $p, p+1, p+2...$
Differences	$\Delta^2 y_{-1}, \Delta^3 y_{-1}$	$\Delta^2 y_{-1}, \Delta^3 y_{-2}$

3.7 Complete Summary

Difference Operators Summary

Operator	Symbol	Formula	Relationship
Forward	Δ	$f(x+h) - f(x)$	$\Delta = E - 1$
Backward	∇	$f(x) - f(x-h)$	$\nabla = 1 - E^{-1}$
Shift	E	$f(x+h)$	$E = 1 + \Delta$

Interpolation Formulas Summary

Method	Formula	Best For
Newton Forward	$y = y_0 + p\Delta y_0 + p(p-1)/2 \Delta^2 y_0 + \dots$	Beginning of data
Newton Backward	$y = y_n + p\nabla y_n + p(p+1)/2 \nabla^2 y_n + \dots$	End of data
Gauss Forward	$y = y_0 + p\Delta y_0 + p(p-1)/2 \Delta^2 y_{-1} + \dots$	Middle (after center)
Gauss Backward	$y = y_0 + p\Delta y_{-1} + (p+1)p/2 \Delta^2 y_{-1} + \dots$	Middle (before center)

Key Relationships

$$\Delta y_i = y_{i+1} - y_i$$

$$\nabla y_i = y_i - y_{i-1}$$

$$\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i$$

$$\Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$$

Important Notes

- Polynomial degree = Order of constant difference
- Higher differences for polynomial are zero
- Choice of method depends on position of interpolation point
- h is the step size ($x_{i+1} - x_i$)

3.8 Previous Year Questions

Short Questions (2 marks)

1. Define finite difference.
2. What is the forward difference operator?
3. What is the relationship between Δ and E ?
4. Define interpolation.
5. When is Newton Forward formula used?
6. What is p in interpolation formulas?
7. What is the shift operator?
8. What is the difference between interpolation and extrapolation?

Medium Questions (5 marks)

1. Construct the difference table for $y = x^3$ at $x = 0, 1, 2, 3, 4$.
2. Find y at $x = 1.4$ using Newton Forward formula from the data:
 $x: 0, 1, 2, 3 \quad y: 1, 4, 9, 16$
3. Explain Gauss Forward and Backward interpolation formulas with examples.
4. Find y at $x = 0.5$ using appropriate interpolation from:
 $x: 0, 1, 2, 3 \quad y: 2, 3, 6, 11$

Long Questions (12 marks)

1. (a) Explain the Newton Forward Interpolation formula with algorithm and example.
(b) From the given data find y at $x = 2.5$:
 $x: 1, 2, 3, 4, 5 \quad y: 2, 5, 10, 17, 26$

2. (a) What are finite differences? Explain different operators (Δ , ∇ , E) with their relationships.
- (b) Construct difference table for $y = x^3$ at $x = 0, 1, 2, 3, 4$ and find y at $x = 1.5$.
3. (a) Explain Gauss Forward and Backward interpolation formulas.
- (b) Find y at $x = 2.5$ and $x = 1.5$ from the data:
- $x: 0, 1, 2, 3, 4$ $y: 1, 8, 27, 64, 125$

Quick Reference

When to Use	Method
Beginning of data	Newton Forward
End of data	Newton Backward
Middle (after center)	Gauss Forward
Middle (before center)	Gauss Backward

Concept	Key Formula
$\Delta = E - 1$	Forward Difference
$\nabla = 1 - E^{-1}$	Backward Difference
$E = 1 + \Delta$	Shift Operator
$p = (x - x_0)/h$	Forward/Gauss Forward
$p = (x - x_n)/h$	Backward